

GANESH GUPTA

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Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Provisioned and operated multi-cloud GPU clusters (AWS, GCP, CoreWeave) with NVIDIA/CUDA hardware and InfiniBand/RoCE networking, benchmarking and tuning placement for long-running training and high-availability LLM inference on shared fleet capacity.
- Wrote infrastructure and systems-level code in Go, Rust, and C++ for GPU scheduling, resource allocation, and fault-tolerant placement logic, packing workloads efficiently to reduce hardware footprint under real GPU-scarcity constraints.
- Built observability and autoscaling for GPU inference serving using Prometheus and Grafana, and ran distributed training via Slurm-based HPC scheduling, keeping fleet utilization and latency SLOs stable through node loss.

QureAI

Jan 2024 – May 2025

ML Infrastructure Engineer

- Cut p99 LLM inference latency 420ms to 140ms across 50K+ daily requests by profiling and migrating synchronous PyTorch-backed serving to async vLLM, optimizing GPU throughput under autoscaling.
- Built a model performance and observability framework tracking latency, drift, and evaluation regressions across production model versions.

Nomura Research Institute Financial Technologies

Aug 2020 – Aug 2023

Software Engineer

- Built and operated 25+ production C++ and SQL-backed distributed systems for 1M+ users, sustaining 99.9% uptime over 3 years on AWS (EC2, RDS, SQS).
- Cut nightly financial data aggregation blocking downstream settlement from 8 minutes to 90 seconds by redesigning SQL with indexed views and batch partitioning.
- Standardized failure-handling contracts across 25+ services, reducing production escalations 60% in one quarter via consistent retry, timeout, and propagation semantics.

Projects

Distributed KV Store / VPC Control Plane Simulator | github.com/ganeshhgupta/distributed-kv-store

- Built a Raft-consensus key-value store in-house with leader election, log replication, and log compaction; companion VPC control plane simulator modeling subnet routing and security group enforcement for fault-tolerant distributed coordination.

Watch-your-LLM | github.com/ganeshhgupta/watch-your-llm

- Production performance and eval tooling with rubric-based scoring, drift detection, and per-call latency benchmarking across model versions; FastAPI backend, PostgreSQL.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: C++, Go, Rust, Python, Java, SQL

GPU & HPC: NVIDIA/CUDA, InfiniBand/RoCE, Multi-Cloud GPU Orchestration (AWS, GCP, CoreWeave), Slurm, PyTorch, vLLM, Inference Latency Optimization

Systems & Observability: Fault Tolerance, Autoscaling, Distributed Systems, Consensus Protocols (Raft), Prometheus, Grafana

Cloud & Infra: AWS (EC2, Lambda, S3), GCP (Cloud Run), Docker, Kubernetes, CI/CD

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer